

Unit 3: Project Monitoring

Experiments Covered in This Unit

Exp 3.1: Project Performance Analysis - Earned Value Management (EVA)

Exp 3.2: Risk Management and Mitigation Planning

Exp 3.3: RMMM Plan, Configuration Management and GitHub

Exp 3.4: Unit Testing with Test Cases

Earned Value Analysis (EVA)

Month 2 Performance Snapshot

Planned Value (PV)

INR 4.20 L

Earned Value (EV)

INR 3.60 L

Actual Cost (AC)

INR 4.50 L

Budget (BAC)

INR 10.00 L

Variance Metrics

- $CV = EV - AC = 3.60 - 4.50 = -0.90 \text{ L}$ (Over Budget)
- $SV = EV - PV = 3.60 - 4.20 = -0.60 \text{ L}$ (Behind Schedule)
- $CPI = EV/AC = 3.60/4.50 = 0.80$ (Every INR 1 spent = INR 0.80 earned)
- $SPI = EV/PV = 3.60/4.20 = 0.857$ (86% schedule efficiency)

Forecasting

- $EAC = BAC/CPI = 10/0.80 = \text{INR } 12.50 \text{ L}$
- $ETC = EAC - AC = 12.50 - 4.50 = \text{INR } 8.00 \text{ L}$
- $TCPI = (BAC - EV)/(BAC - AC) = 6.40/5.50 = 1.164$
- Need 16.4% higher efficiency on remaining work
- Corrective: API batch mode + task parallelisation

Risk Management & Mitigation

6 Identified Risks with Response Strategies

Risk ID	Risk Description	Probability	Impact	Score	Strategy
R01	Gemini API rate limits under load	0.60	0.80	0.48	Mitigate - API caching + batch mode
R02	AI module taking longer than 18.67 days	0.50	0.80	0.40	Mitigate - Sprint sub-tasks + fallback static
R03	Firebase costs exceeding budget	0.50	0.50	0.25	Mitigate - Spark tier + billing alerts
R04	Abacus UI development delays	0.40	0.50	0.20	Mitigate - Use react-abacus library base
R05	Low student engagement / adoption	0.30	0.60	0.18	Accept + UAT with 5 students pre-launch
R06	Team availability disruption	0.30	0.70	0.21	Mitigate - 2-day sprint buffers + PM backup

Red = High Risk (score $\geq$ 0.40) | Blue = Medium Risk | Contingency Reserve: INR 2.00 Lakhs total

Configuration Management & GitHub Strategy

Git Flow + CI/CD + Versioning

Git Flow Branching Strategy

- main: Production-ready code | Deployed to Vercel
- develop: Integration branch | 1 approval required
- feature/*: Individual module branches | Deleted after merge
- release/*: Release preparation + final bug fixes
- hotfix/*: Urgent production fixes
- Semantic Versioning: v0.1.0 (alpha) → v1.0.0 (production)

CI/CD Pipeline Stages

- Stage 1: ESLint + Prettier - Lint check on every push
- Stage 2: Jest unit tests (min 80% coverage) on PR
- Stage 3: Next.js production build verification
- Stage 4: Vercel Preview Deploy on PR to develop
- Stage 5: Vercel Production Deploy on merge to main
- Environment vars stored in GitHub Secrets (never in code)
- Docker: Reproducible dev environment across all machines

Unit Testing & Quality Dashboard

Sprint 2 Test Execution Summary

Module	Test Cases	Passed	Failed	Pass Rate
Abacus Simulator	20	18	2	90.0%
Number System Explorer	15	14	1	93.3%
AI Story Module	14	10	4	71.4%
Quiz Module	14	12	2	85.7%
Auth & Access	9	8	1	88.9%
Integration	7	6	1	85.7%
TOTAL	79	68	11	86.1%

Key Defects Found

- CRITICAL: AI response time 8.3s (SLA breach)
- CRITICAL: Gemini cold-start latency on idle
- MAJOR: Abacus carry-over logic error (>9)
- MINOR: Mobile UI alignment in quiz module
- Fix: Vercel cron warm-up every 4 minutes
- Fix: React useReducer for bead state mgmt
- Target Pass Rate: 95% | Final: 96.2%